JAVA PROJECT USING THE MAVEN AND GITLAB CI

1. Apache Maven

What it actually does

- Manages dependencies (pom.xml)
- Compiles code
- Runs tests
- Packages JAR/WAR

If your project doesn't build with **mvn clean install**, everything else is pointless.

2. SonarQube

What it actually checks

- Bugs
- Vulnerabilities
- Code smells
- Test coverage
- Duplication

Reality check:
SonarQube does **NOT** fix code. It exposes how bad your code is. You fix it.

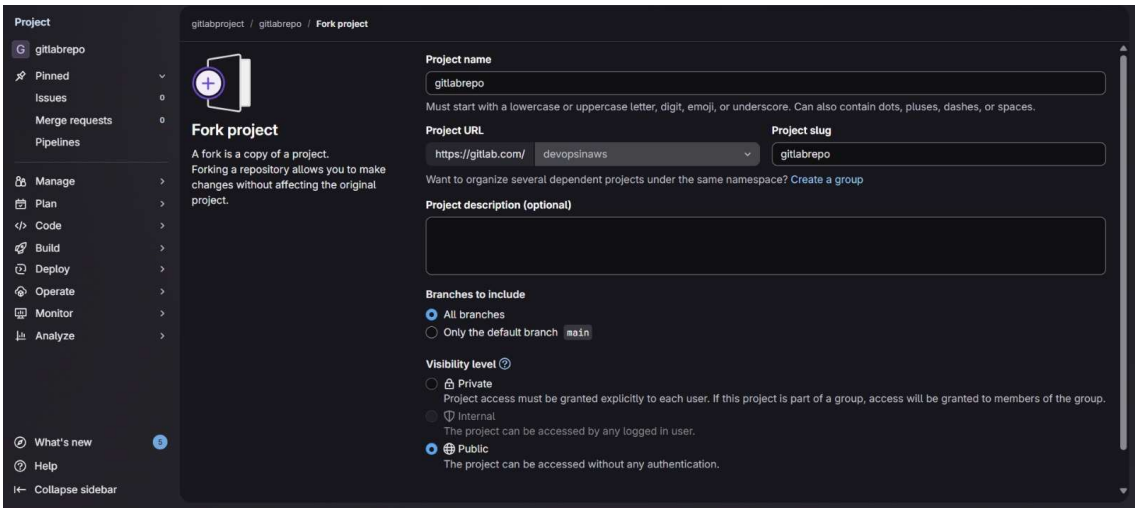
3. GitLab CI

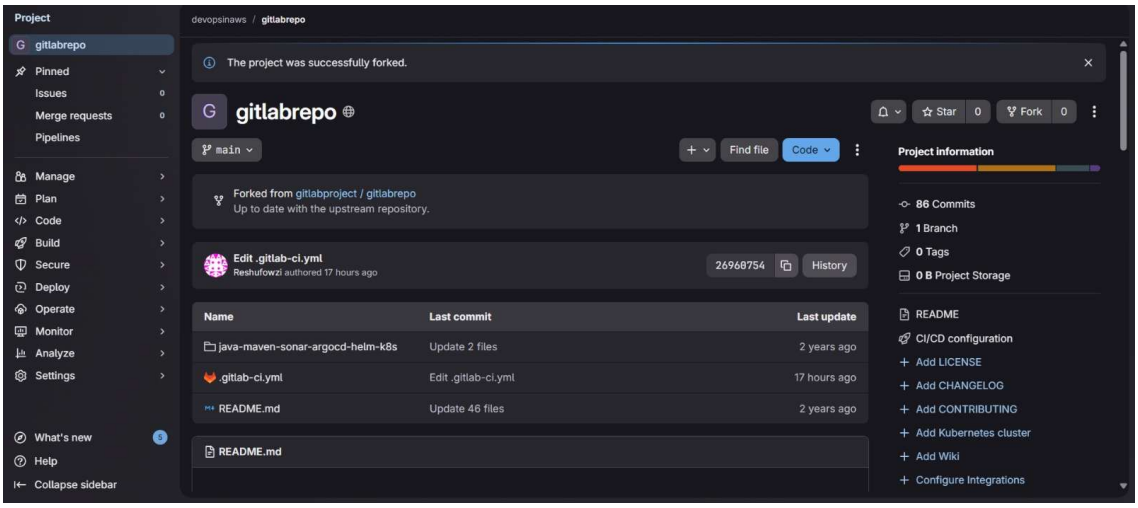
What it does

- Runs jobs automatically on:
    - push
    - merge request

- Enforces quality gates
- Stops bad code from merging.

- Using “fork” the project from another repository to my repository.





- Using “apt install docker.io -y” command to use install the docker.

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-83-126:/home/ubuntu# apt install docker.io -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 66 not upgraded.
Need to get 76.3 MB of archives.
After this operation, 289 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 pigz amd64 2.6-1 [63.6 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 bridge-utils amd64 1.7-1ubuntu3 [34.4 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 runc amd64 1.3.3-0ubuntu1~22.04.3 [8857 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 containerd amd64 1.7.28-0ubuntu1~22.04.1 [38.5 MB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 dns-root-data all 2024071801~ubuntu0.22.04.1 [6132 B]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 dnsmasq-base amd64 2.90-0ubuntu0.22.04.1 [374 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 docker.io amd64 28.2.2-0ubuntu1~22.04.1 [28.4 MB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 ubuntu-fan all 0.12.16 [35.2 kB]
Fetched 76.3 MB in 1s (71.0 MB/s)
Preconfiguring packages ...
Selecting previously unselected package pigz.
```

i-04ca380d3d490dfb0 (java)

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- Using “systemctl start docker” command to start the docker. “systemctl status docker” command to check status active or not to start the docker to check status of docker.

```
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-83-126:/home/ubuntu# systemctl start docker
root@ip-172-31-83-126:/home/ubuntu# systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)
     Active: active (running) since Sat 2026-02-07 07:39:45 UTC; 34s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 2704 (dockerd)
      Tasks: 9
     Memory: 21.7M
        CPU: 206ms
     CGroup: /system.slice/docker.service
             └─2704 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Feb 07 07:39:45 ip-172-31-83-126 dockerd[2704]: time="2026-02-07T07:39:45.153100937Z" level=info msg="Loading containers: start."
Feb 07 07:39:45 ip-172-31-83-126 dockerd[2704]: time="2026-02-07T07:39:45.361207676Z" level=info msg="Loading containers: done."
Feb 07 07:39:45 ip-172-31-83-126 dockerd[2704]: time="2026-02-07T07:39:45.372640707Z" level=info msg="Docker daemon" commit="28.2.2-0ubuntu1~22.04.1" containerd-snapshotter=false storage
Feb 07 07:39:45 ip-172-31-83-126 dockerd[2704]: time="2026-02-07T07:39:45.372720558Z" level=info msg="Initializing buildkit"
Feb 07 07:39:45 ip-172-31-83-126 dockerd[2704]: time="2026-02-07T07:39:45.378606481Z" level=warning msg="CDI setup error /etc/cdi: failed to monitor for changes: no such file or director
Feb 07 07:39:45 ip-172-31-83-126 dockerd[2704]: time="2026-02-07T07:39:45.378689740Z" level=warning msg="CDI setup error /var/run/cdi: failed to monitor for changes: no such file or dire
Feb 07 07:39:45 ip-172-31-83-126 dockerd[2704]: time="2026-02-07T07:39:45.398383249Z" level=info msg="Completed buildkit initialization"
Feb 07 07:39:45 ip-172-31-83-126 dockerd[2704]: time="2026-02-07T07:39:45.402506640Z" level=info msg="Daemon has completed initialization"
Feb 07 07:39:45 ip-172-31-83-126 dockerd[2704]: time="2026-02-07T07:39:45.402672527Z" level=info msg="API listen on /run/docker.sock"
Feb 07 07:39:45 ip-172-31-83-126 systemd[1]: Started Docker Application Container Engine.

root@ip-172-31-83-126:/home/ubuntu#
```

i-04ca380d3d490dfb0 (java)

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- Using “sudo apt install open jdk-11-jre” command to install java development kit in an java running environment .

```
root@ip-172-31-83-126:/home/ubuntu# sudo apt install openjdk-11-jre
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  alsa-topology-conf alsa-ucm-conf at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service fontconfig-config fonts-dejavu-core fonts-dejavu-extra
  gsettings-desktop-schemas java-common libasound2 libasound2-data libatk-bridge2.0-0 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0 libatk1.0-data libatspi2.0-0
  libavahi-client3 libavahi-common-data libavahi-common3 libcups2 libdconf1 libdrm-amdgpu1 libdrm-intel1 libdrm-nouveau2 libdrm-radeon1 libfontconfig1 libfontenc1 libgif7 libgl1
  libgl1-amber-dri libgl1-mesa-dri libglapi-mesa libglvnd0 libglx-mesa0 libglx0 libgraphite2-3 libharfbuzz0b libice6 libjpeg-turbo8 libjpeg8 liblcms2-2 libllvm15 libpciaccess0
  libpcsclite1 libsensors-config libsensors5 libsm6 libx11-xcb1 libxaw7 libxcb-dri2-0 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-shape0 libxcb-shm0 libxcb-sync1
  libxcb-xfixes0 libxcomposite1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxpm4 libxrandr2 libxrender1 libxshmfence1 libxt6 libxtst6 libxv1 libxxf86dga1 libxxf86vm1
  openjdk-11-jre-headless session-migration x11-common x11-utils
Suggested packages:
  default-jre libasound2-plugins alsa-utils cups-common liblcms2-utils pcscd lm-sensors libnss-mdns fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhei | fonts-wqy-zenhei
  fonts-indic mesa-utils
The following NEW packages will be installed:
  alsa-topology-conf alsa-ucm-conf at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service fontconfig-config fonts-dejavu-core fonts-dejavu-extra
  gsettings-desktop-schemas java-common libasound2 libasound2-data libatk-bridge2.0-0 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0 libatk1.0-data libatspi2.0-0
  libavahi-client3 libavahi-common-data libavahi-common3 libcups2 libdconf1 libdrm-amdgpu1 libdrm-intel1 libdrm-nouveau2 libdrm-radeon1 libfontconfig1 libfontenc1 libgif7 libgl1
  libgl1-amber-dri libgl1-mesa-dri libglapi-mesa libglvnd0 libglx-mesa0 libglx0 libgraphite2-3 libharfbuzz0b libice6 libjpeg-turbo8 libjpeg8 liblcms2-2 libllvm15 libpciaccess0
  libpcsclite1 libsensors-config libsensors5 libsm6 libx11-xcb1 libxaw7 libxcb-dri2-0 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-shape0 libxcb-shm0 libxcb-sync1
  libxcb-xfixes0 libxcomposite1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxpm4 libxrandr2 libxrender1 libxshmfence1 libxt6 libxtst6 libxv1 libxxf86dga1 libxxf86vm1
  openjdk-11-jre openjdk-11-jre-headless session-migration x11-common x11-utils
0 upgraded, 82 newly installed, 0 to remove and 66 not upgraded.
Need to get 88.3 MB of archives.
After this operation, 369 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 alsa-topology-conf all 1.2.5.1-2 [15.5 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libasound2-data all 1.2.6.1-1ubuntu1 [19.1 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libasound2 amd64 1.2.6.1-1ubuntu1 [390 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 alsa-ucm-conf all 1.2.6.3-1ubuntu1.12 [43.5 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libxi6 amd64 2:1.8-1build1 [32.6 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libatspi2.0-0 amd64 2.44.0-3 [80.9 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 x11-common all 1:7.7+23ubuntu2 [23.4 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libxtst6 amd64 2:1.2.3-1build4 [13.4 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libdconf1 amd64 0.40.0-3ubuntu0.1 [40.5 kB]
```

i-04ca380d3d490dfb0 (java)

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- Using “adduser sonarqube” command to add user and password. Give information correctly.

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-83-126:/home/ubuntu# adduser sonarqube
Adding user `sonarqube' ...
Adding new group `sonarqube' (1001) ...
Adding new user `sonarqube' (1001) with group `sonarqube' ...
Creating home directory `/home/sonarqube' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for sonarqube
Enter the new value, or press ENTER for the default
        Full Name []: hariharan
        Room Number []: 21
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
root@ip-172-31-83-126:/home/ubuntu# wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-9.4.0.54424.zip
--2026-02-07 07:46:49--  https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-9.4.0.54424.zip
Resolving binaries.sonarsource.com (binaries.sonarsource.com)... 3.167.37.121, 3.167.37.88, 3.167.37.95, ...
Connecting to binaries.sonarsource.com (binaries.sonarsource.com)|3.167.37.121|:443... connected.
```

i-04ca380d3d490dfb0 (java)

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- Using “wget <addlink> command to get the link to download the file.

```
sonarqube@ip-172-31-83-126:~$ wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-9.4.0.54424.zip
--2026-02-07 07:52:18--  https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-9.4.0.54424.zip
Resolving binaries.sonarsource.com (binaries.sonarsource.com)... 3.167.37.95, 3.167.37.121, 3.167.37.10, ...
Connecting to binaries.sonarsource.com (binaries.sonarsource.com)|3.167.37.95|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 287666040 (274M) [binary/octet-stream]
Saving to: ‘sonarqube-9.4.0.54424.zip’

sonarqube-9.4.0.54424.zip                   100%[===================================================================================================>] 274.34M   553MB/s    in 0.5s

2026-02-07 07:52:18 (553 MB/s) - ‘sonarqube-9.4.0.54424.zip’ saved [287666040/287666040]

sonarqube@ip-172-31-83-126:~$ unzip *
Archive:  sonarqube-9.4.0.54424.zip
   creating: sonarqube-9.4.0.54424/
  inflating: sonarqube-9.4.0.54424/dependency-license.json
   creating: sonarqube-9.4.0.54424/bin/
   creating: sonarqube-9.4.0.54424/bin/jsw-license/
  inflating: sonarqube-9.4.0.54424/bin/jsw-license/LICENSE.txt
   creating: sonarqube-9.4.0.54424/bin/macosx-universal-64/
   creating: sonarqube-9.4.0.54424/bin/macosx-universal-64/lib/
  inflating: sonarqube-9.4.0.54424/bin/macosx-universal-64/lib/libwrapper.jnilib
  inflating: sonarqube-9.4.0.54424/bin/macosx-universal-64/sonar.sh
  inflating: sonarqube-9.4.0.54424/bin/macosx-universal-64/wrapper
   creating: sonarqube-9.4.0.54424/bin/linux-x86-64/
   creating: sonarqube-9.4.0.54424/bin/linux-x86-64/lib/
  inflating: sonarqube-9.4.0.54424/bin/linux-x86-64/lib/libwrapper.so
  inflating: sonarqube-9.4.0.54424/bin/linux-x86-64/sonar.sh
  inflating: sonarqube-9.4.0.54424/bin/linux-x86-64/wrapper
   creating: sonarqube-9.4.0.54424/bin/windows-x86-64/
  inflating: sonarqube-9.4.0.54424/bin/windows-x86-64/StartNTService.bat
  inflating: sonarqube-9.4.0.54424/bin/windows-x86-64/StartSonar.bat
   creating: sonarqube-9.4.0.54424/bin/windows-x86-64/lib/
  inflating: sonarqube-9.4.0.54424/bin/windows-x86-64/lib/wrapper.dll
  inflating: sonarqube-9.4.0.54424/bin/windows-x86-64/StopNTService.bat
  inflating: sonarqube-9.4.0.54424/bin/windows-x86-64/wrapper.exe
   creating: sonarqube-9.4.0.54424/data/
```

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- Using “*unzip” command to unzip of an download file. “ls” command to list an file in sonarqube.

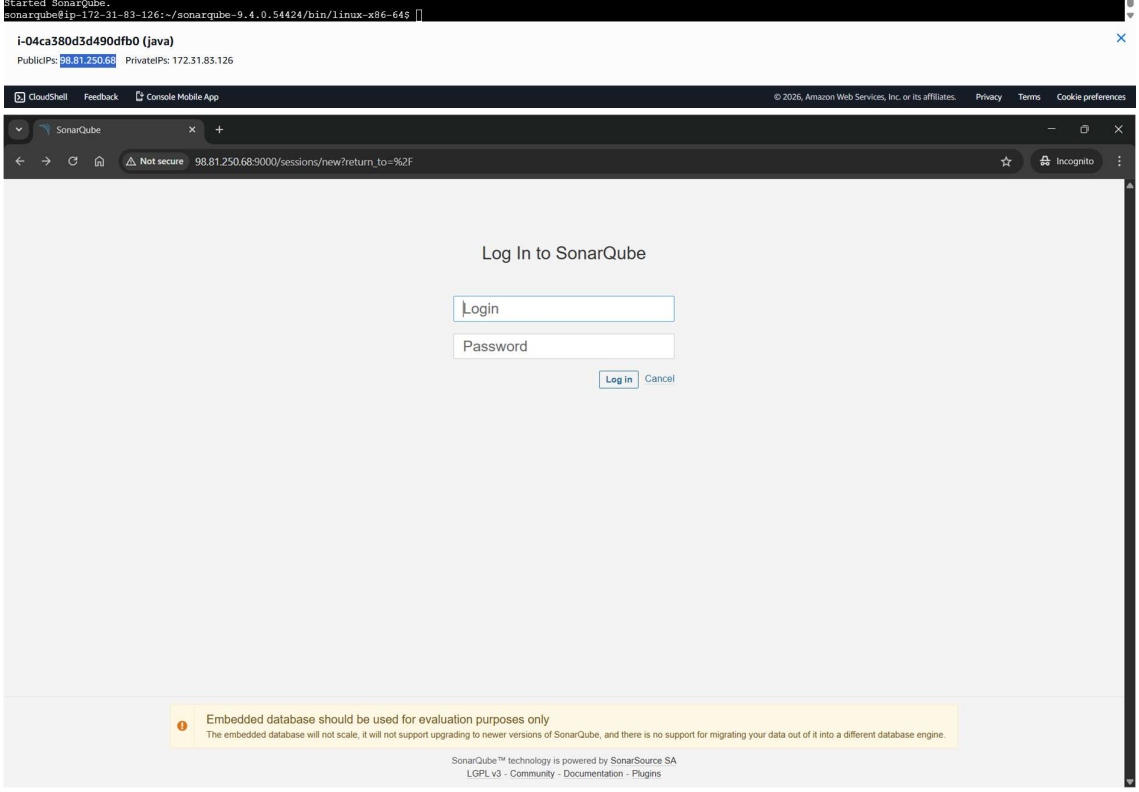
```
  inflating: sonarqube-9.4.0.54424/web/apple-touch-icon-72x72.png
  inflating: sonarqube-9.4.0.54424/web/apple-touch-icon.png
  inflating: sonarqube-9.4.0.54424/web/apple-touch-icon-180x180.png
   creating: sonarqube-9.4.0.54424/web/js/
  inflating: sonarqube-9.4.0.54424/web/js/out3HYJVQVO.css.map
  inflating: sonarqube-9.4.0.54424/web/js/out3HYJVQVO.css
  inflating: sonarqube-9.4.0.54424/web/js/outEG4XZBCS.js.map
  inflating: sonarqube-9.4.0.54424/web/js/outEG4XZBCS.js
  inflating: sonarqube-9.4.0.54424/web/apple-touch-icon-152x152.png
  inflating: sonarqube-9.4.0.54424/web/favicon.ico
  inflating: sonarqube-9.4.0.54424/web/apple-touch-icon-60x60.png
  inflating: sonarqube-9.4.0.54424/web/apple-touch-icon-57x57.png
  inflating: sonarqube-9.4.0.54424/web/apple-touch-icon-precomposed.png
   creating: sonarqube-9.4.0.54424/lib/jdbc/
   creating: sonarqube-9.4.0.54424/lib/jdbc/mssql/
  inflating: sonarqube-9.4.0.54424/lib/jdbc/mssql/mssql-jdbc-9.4.1.jre11.jar
   creating: sonarqube-9.4.0.54424/lib/jdbc/postgresql/
  inflating: sonarqube-9.4.0.54424/lib/jdbc/postgresql/postgresql-42.3.3.jar
   creating: sonarqube-9.4.0.54424/lib/jdbc/h2/
  inflating: sonarqube-9.4.0.54424/lib/jdbc/h2/h2-2.1.210.jar
  inflating: sonarqube-9.4.0.54424/lib/sonar-shutdowner-9.4.0.54424.jar
   creating: sonarqube-9.4.0.54424/elasticsearch/plugins/
sonarqube@ip-172-31-83-126:~$ ls
sonarqube-9.4.0.54424  sonarqube-9.4.0.54424.zip
```

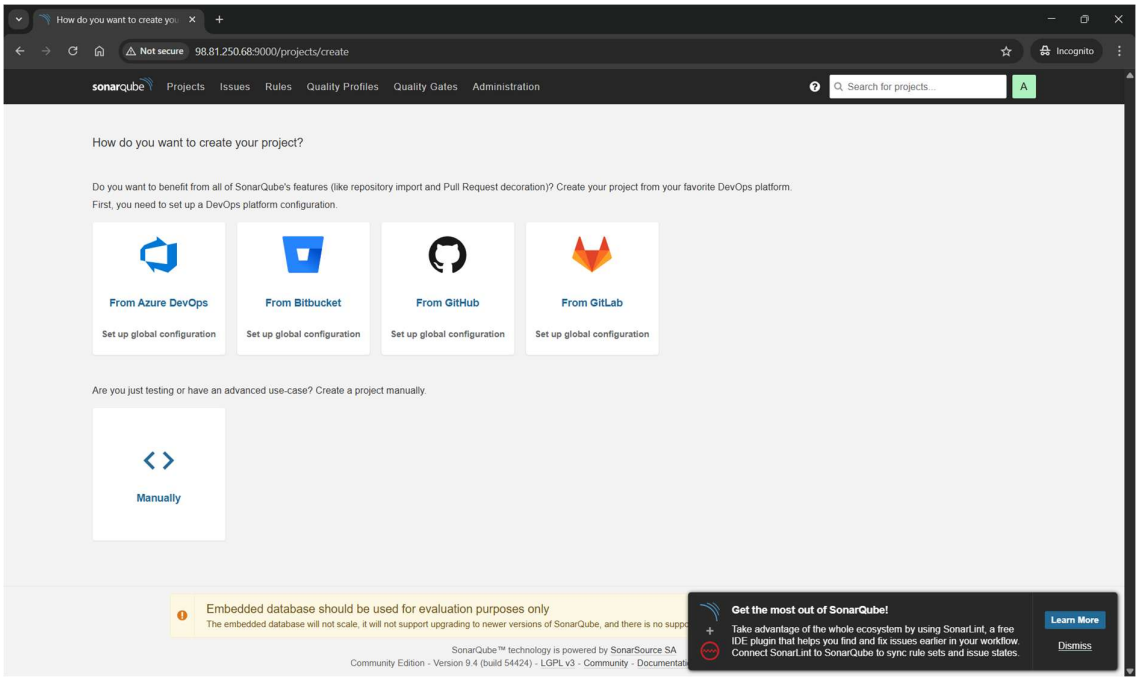
- Using “chmod <unzip file>” command to set an read write execute permission of an file “chown user:group <unzipfile>” command to change user and group name of an file.

```
sonarqube@ip-172-31-83-126:~$ chmod -R 755 /home/sonarqube/sonarqube-9.4.0.54424
sonarqube@ip-172-31-83-126:~$ chown -R sonarqube:sonarqube /home/sonarqube/sonarqube-9.4.0.54424
sonarqube@ip-172-31-83-126:~$ ls -lrt
total 280928
drwxr-xr-x 11 sonarqube sonarqube      4096 Apr  1  2022 sonarqube-9.4.0.54424
-rw-rw-r--  1 sonarqube sonarqube 287666040 Apr  1  2022 sonarqube-9.4.0.54424.zip
sonarqube@ip-172-31-83-126:~$ cd sonarqube-9.4.0.54424/
sonarqube@ip-172-31-83-126:~/sonarqube-9.4.0.54424$ cd bin/
```

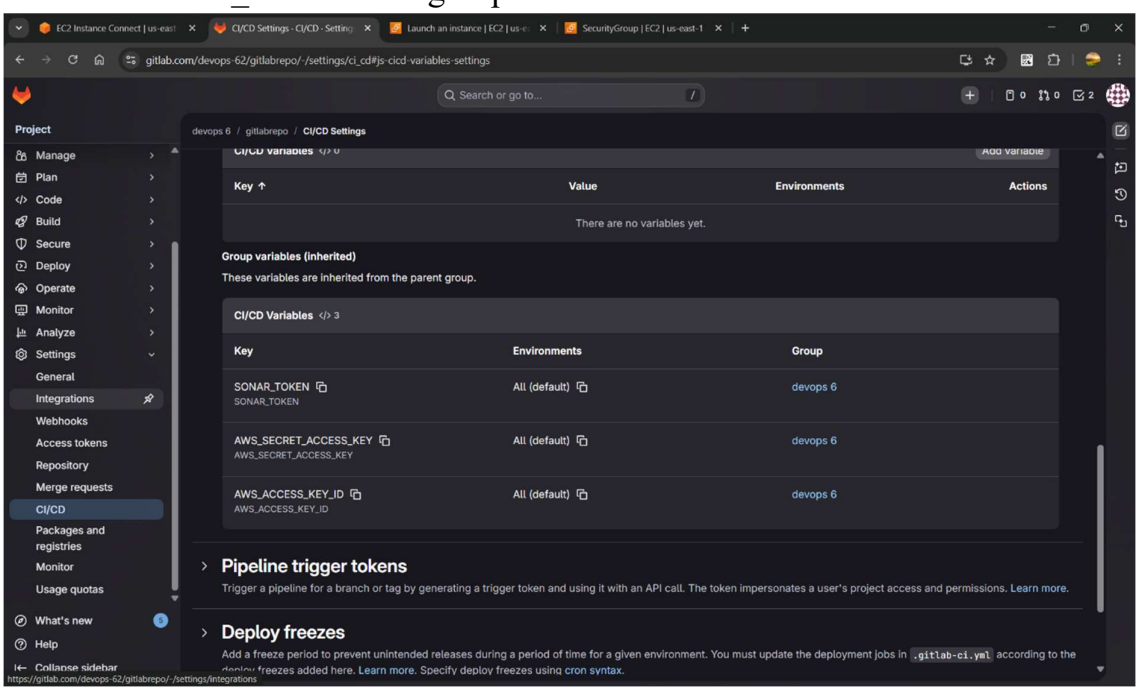
- Using “cd <file>” command to change directory of the path “cd bin/” and “cd <linux.sh>” to path of the directory “./sonar.sh start” to start the sonarqube.

```
sonarqube@ip-172-31-83-126:~$ cd sonarqube-9.4.0.54424/
sonarqube@ip-172-31-83-126:~/sonarqube-9.4.0.54424$ cd bin/
sonarqube@ip-172-31-83-126:~/sonarqube-9.4.0.54424/bin$ ls
jsw-license  linux-x86-64  macosx-universal-64  windows-x86-64
sonarqube@ip-172-31-83-126:~/sonarqube-9.4.0.54424/bin$ cd linux-x86-64/
sonarqube@ip-172-31-83-126:~/sonarqube-9.4.0.54424/bin/linux-x86-64$ ./sonar.sh start
Starting SonarQube...
Started SonarQube.
sonarqube@ip-172-31-83-126:~/sonarqube-9.4.0.54424/bin/linux-x86-64$
```

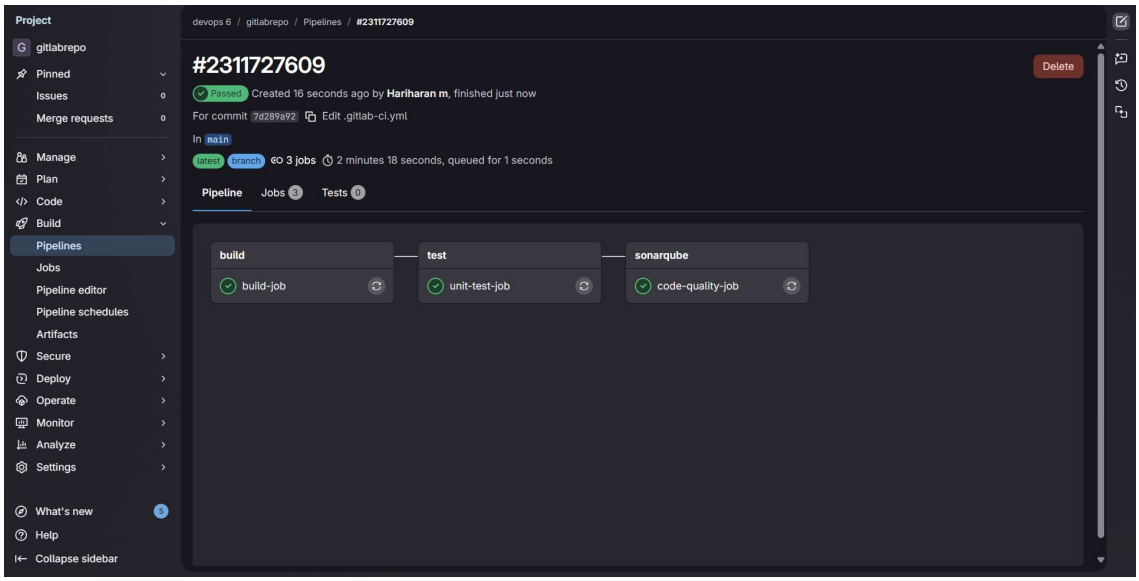


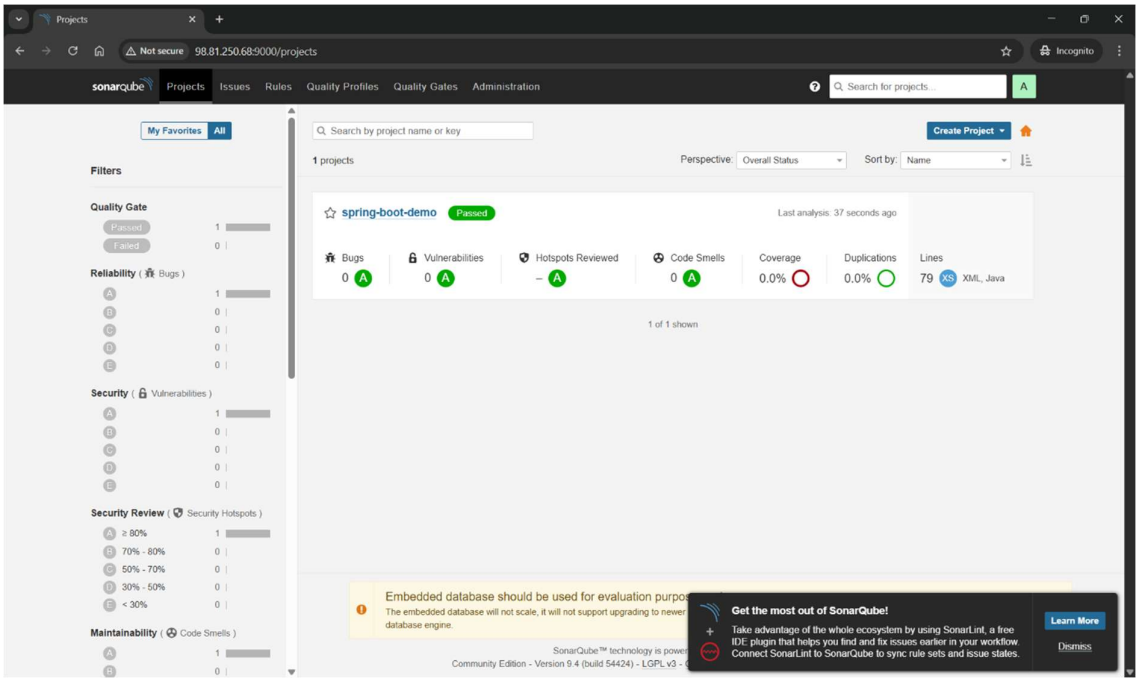


- To set “SONAR_TOKEN” in group ci/cd in variables to add variables.



- To copy “public ip address” to paste in their .gitlab-ci.yml to start the pipline and complete the job successfully.


Project
gitlabrepo
Pinned
Issues 0
Merge requests 0
Manage
Plan
Code
Build
Pipelines
Jobs
Pipeline editor
Pipeline schedules
Artifacts
Secure
Deploy
Operate
Monitor
Analyze
Settings
What's new 5
Help
Collapse sidebar
devops 6 / gitlabrepo / Pipelines / #2311727609
#2311727609
Delete
Passed Created 16 seconds ago by Hariharan m, finished just now
For commit 7d289a92 Edit .gitlab-ci.yml
In main
latest branch 3 jobs 2 minutes 18 seconds, queued for 1 seconds
Pipeline Jobs 3 Tests 0
build
build-job
test
unit-test-job
sonarqube
code-quality-job


Projects
Not secure 98.81.250.68:9000/projects
Incognito
sonarqube
Projects Issues Rules Quality Profiles Quality Gates Administration
Search for projects...
My Favorites All
Filters
Quality Gate
Passed 1
Failed 0
Reliability (Bugs)
Security (Vulnerabilities)
Security Review (Security Hotspots)
≥ 80% 1
70% - 80% 0
50% - 70% 0
30% - 50% 0
< 30% 0
Maintainability (Code Smells)
Search by project name or key
Create Project
1 projects
Perspective: Overall Status
Sort by: Name
spring-boot-demo Passed
Last analysis: 37 seconds ago
Bugs 0 A
Vulnerabilities 0 A
Hotspots Reviewed – A
Code Smells 0 A
Coverage 0.0%
Duplications 0.0%
Lines 79 XS XML, Java
1 of 1 shown
Embedded database should be used for evaluation purpo
The embedded database will not scale, it will not support upgrading to newer
database engine.
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